



9545 - The Substellar Missing Link: Brown Dwarfs in Globular Clusters

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Thomas M. Brown (PI)	Space Telescope Science Institute
Roberto Avila (CoI)	Space Telescope Science Institute
Dr. Matteo Correnti (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Mario Gennaro (CoI)	Space Telescope Science Institute
Dr. Elena Manjavacas (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Dr. Pier Giorgio Prada Moroni (CoI) (ESA Member)	Universita di Pisa
Dr. I. Neill Reid (CoI)	Space Telescope Science Institute

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NGC6397 Observations				
	1		NIRCam Imaging	(1) NGC6397
M4 Observations				
	2		NIRCam Imaging	(2) M4

ABSTRACT

Brown dwarfs inhabit the last unexplored territory in the HR diagram of globular clusters. For the nearest globular clusters, we have a wealth of data for all phases of stellar evolution between the H-burning main sequence and the termination of the white-dwarf cooling curve. However, to date we have only a few candidate brown dwarfs in the nearest of these primeval systems. Our observational knowledge for substellar objects is dominated by field samples in the immediate solar neighborhood, young star-forming regions, and open clusters, where the populations are overwhelmingly skewed toward high metallicities and young ages.

JWST Proposal 9545 (Created: Friday, March 13, 2026, 12:00:58PM Eastern Standard Time) - Overview

We propose to observe the nearest two globular clusters, M4 and NGC6397, to depths unambiguously below the H-burning limit, reaching brown dwarfs of 0.063 to 0.070 solar masses. This range includes L0-L9 subdwarfs, the L/T transition (L8-T4), and T0-T6 subdwarfs in each cluster. The L/T transition is poorly constrained at old ages and low metallicities, given the characteristics of the local stellar population and the complexities of subdwarf model atmospheres. The reduced condensate fraction in metal-poor subdwarf atmospheres could influence the L/T transition in this regime, with distinct cloud compositions and dynamics from those in their solar-metallicity counterparts. Targeting well-studied stellar clusters will yield subdwarf samples with high-precision ages, chemical abundances, and distances. The resulting constraints on their physical parameters (mass, temperature, luminosity), combined with insight into the formation efficiencies of these objects, will anchor substellar models in the ancient metal-poor regime.

OBSERVING DESCRIPTION

We require $\text{SNR} \geq 10$ photometry per band, within a 2-pix radius, for a mass of 0.063 M_{sun} in M4 ($m_{150} = 28.3$ mag, $m_{322} = 25.5$ mag), and 0.070 M_{sun} in NGC6397 ($m_{150} = 29.0$ mag, $m_{322} = 26.1$ mag), reaching T6 subdwarfs below 1000K in each GC. For T6 subdwarfs, F150W2 drives the time to reach $\text{SNR} \geq 10$ ($\text{SNR} > 20$ for F322W2). SNR increases with BD mass, with $\text{SNR} > 100$ at the HBL. Both GCs are well above the Galactic plane, with good contrast against the local disk, but we will use PMs to confirm GC members. With a 7-month baseline between JWST/NIRCam observations of 47Tuc, Scalco+(2025) measured sufficiently accurate PMs to confirm members; here, our baseline will be $> 6\times$ larger, and the PMs will be 3 to 4x larger (> 0.5 pix/yr in F150W2), making the measurements much easier. The JWST Exposure Time Calculator (ETC) confirms that F150W2 and F322W2 are the most efficient bands for BDs in this mass range. We used BD models with synthetic photometry calculated in these bands to normalize our ETC calculations, avoiding systematic errors from using ETC spectra inappropriate for BDs. Note that this program is impossible with HST, requiring an exposure time exceeding that available in an HST observing cycle. Shallower Cycle 1 images of both GCs (GO-1979) are sufficient to provide first-epoch astrometry and PMs for all objects of interest here.

Following JDox advice, we specify > 4 groups per integration and generous dithers to sample the point-spread function and mitigate artifacts. For NGC6397, we use the MEDIUM8 readout, 9 groups/integration, and 30 dithers, for a total exposure time of 28,350 sec. For M4, we use the MEDIUM8 readout, with 6 groups/integration, and 16 dithers, for a total exposure time of 9,964 sec.

Special requirements: Aperture PA Range. All BDs must be confirmed as GC members via PMs, requiring maximum overlap with archival GO-1979 data, and the same V3PAs from Cycle 1: 267.0714 degrees (NGC6397) and 290.0714 degrees (M4). Flipping NGC6397 by 180 degrees modestly increases the background, reducing SNR by 4%, so we add this option. Flipping M4 by 180 degrees is unavailable.

Proposal 9545 - Targets - The Substellar Missing Link: Brown Dwarfs in Globular Clusters

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC6397	RA: 17 41 6.0000 (265.2750000d) Dec: -53 45 20.00 (-53.75556d) Equinox: J2000	Proper Motion RA: 3.714e-4 sec of time/yr Proper Motion Dec: -0.0176 arcsec/yr Epoch of Position: 2015.5	
	Comments: Category=Stellar Cluster Description=[Globular star clusters]				
Fixed Targets	(2)	M4	RA: 16 23 43.2500 (245.9302083d) Dec: -26 27 0.00 (-26.45000d) Equinox: J2000	Proper Motion RA: -9.2989e-4 sec of time/yr Proper Motion Dec: -0.01899 arcsec/yr Epoch of Position: 2015.5	
	Comments: Category=Stellar Cluster Description=[Globular star clusters]				

Proposal 9545 - Observation 1 - The Substellar Missing Link: Brown Dwarfs in Globular Clusters

Observation	Proposal 9545, Observation 1										Fri Mar 13 17:00:58 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCcam Imaging											
	Comments: Target coordinates and V3PA chosen to match exactly the observation position and VAP3 (267.0714) for NGC6397 in GO-1979 (Obs 1), although the GO-1979 observations were shallower (7476 sec science time). We have added a second VAP3 special requirement of 267.0714-180=87.0714 to allow for additional scheduling opportunities and for scheduling outside of the MAZ (267.0714 is within the MAZ but 87.0714 is not).											
Diagnostics	(Observation 1) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	NGC6397	RA: 17 41 6.0000 (265.2750000d) Dec: -53 45 20.00 (-53.75556d) Equinox: J2000				Proper Motion RA: 3.714e-4 sec of time/yr Proper Motion Dec: -0.0176 arcsec/yr Epoch of Position: 2015.5					
	Comments: Category=Stellar Cluster Description=[Globular star clusters]											
Template	Module		Subarray					Target Placement				
	ALL		FULL					Module gap (large extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions		
	1	NONE				STANDARD				30		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F150W2	F322W2	MEDIUM8	9	1	30	30	28345.073			
Special Requirements	Aperture PA Range 86.99682306 to 86.99682306 Degrees (V3 87.0714 to 87.0714) Aperture PA Range 266.99682306 to 266.99682306 Degrees (V3 267.0714 to 267.0714)											

Proposal 9545 - Observation 2 - The Substellar Missing Link: Brown Dwarfs in Globular Clusters

Observation	Proposal 9545, Observation 2										Fri Mar 13 17:00:58 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCam Imaging										
	Comments: Target coordinates and V3PA chosen to match exactly the observation position and PAV3 (290.0714) for M4 in GO-1979 (Obs 2), although the GO-1979 observations were shallower (6192 sec science time). We cannot add a second VAP3 special requirement of 290.0714-180=110.0714 to allow for additional scheduling opportunities, because that VAP3 is unavailable. Note that other VAP3 values throughout the year would misalign our images to archivl images, and also incur extremely high backgrounds that would hamper the science goals.										
	(Observation 2) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
Fixed Targets	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Template											
Dithers											
Spectral Elements											
Special Requirements											